

ADDRESSING MODES OF 8086

8086 provides different addressing modes for Data, Program and Stack Memory.

ADDRESSING MODES FOR DATA MEMORY {IMP}

I IMMEDIATE ADDRESSING MODE

In this mode the **operand** is specified in the **instruction** itself.
Instructions are **longer** but the **operands** are **easily identified**.

Eg: **MOV CL, 12H** ; Moves 12 immediately into CL register
 MOV BX, 1234H ; Moves 1234 immediately into BX register

II REGISTER ADDRESSING MODE

In this mode **operands** are specified using **registers**.
Instructions are **shorter** but **operands cant** be **identified** by looking at the instruction.

Eg: **MOV CL, DL** ; Moves data of DL register into CL register
 MOV AX, BX ; Moves data of BX register into AX register

III DIRECT ADDRESSING MODE

In this mode **address** of the operand is directly specified **in the instruction**.
Here **only** the **offset address is specified**, the segment being indicated by the instruction.

Eg: **MOV CL, [4321H]** ; Moves data from location 4321H in the data
 ; segment into CL
 ; The physical address is calculated as
 ; **DS * 10_H + 4321**
 ; Assume DS = 5000H
 ; ∴ P A= 50000 + 4321 = 54321H
 ; ∴ CL ← [54321H]

Eg: **MOV CX, [4320H]** ; Moves data from location 4320H and 4321H
 ; in the data segment into CL and CH resp.

IV INDIRECT ADDRESSING MODES

REGISTER INDIRECT ADDRESSING MODE

In this mode the μP uses any of the 2 **base registers** BP, BX or any of the two index registers SI, DI to provide the offset **address** for the data byte.

The segment is indicated by the Base Registers:

BX -- Data Segment, BP --- Stack Segment

Eg: MOV CL, [BX] ; Moves a byte from the address pointed by BX in Data
 ; Segment into CL.
 ; Physical Address calculated as $DS * 10_H + BX$

Eg: MOV [BP], CL ; Moves a byte from CL into the location pointed by BP in
 ; Stack Segment.
 ; Physical Address calculated as $SS * 10_H + BP$

REGISTER RELATIVE ADDRESSING MODE

In this mode the operand address is calculated using one of the **base registers** and a **8-bit** or a **16-bit displacement**.

Eg: MOV CL, [BX+4] ; Moves a byte from the address pointed by BX+4 in
 ; Data Seg to CL.
 ; Physical Address: $DS * 10_H + BX + 4H$

Eg: MOV 12H [BP], CL ; Moves a byte from CL to location pointed by BP+12H in
 ; the Stack Seg.
 ; Physical Address: $SS * 10_H + BP + 12H$

BASE INDEXED ADDRESSING MODE

Here, operand address is calculated as **Base register plus** an **Index** register.

Eg: MOV CL, [BX+SI] ; Moves a byte from the address pointed by BX+SI
 ; in Data Segment to CL.
 ; Physical Address: $DS * 10_H + BX + SI$

Eg: MOV [BP+DI], CL ; Moves a byte from CL into the address pointed by
 ; BP+DI in Stack Segment.
 ; Physical Address: $SS * 10_H + BP + DI$

BASE RELATIVE PLUS INDEX ADDRESSING MODE

In this mode the address of the operand is calculated as **Base** register **plus** **Index** register **plus** **8-bit** or **16-bit displacement**.

Eg: MOV CL, [BX+DI+20] ; Moves a byte from the address pointed by
 ; BX+SI+20H in Data Segment to CL.
 ; Physical Address: DS * 10_H + BX + SI+ 20H

Eg: MOV [BP+SI+2000], CL ; Moves a byte from CL into the location pointed by
 ; BP+SI+2000H in Stack Segment.
 ; Physical Address: SS * 10_H + BP+SI+2000H

V IMPLIED ADDRESSING MODE

In this addressing mode the operands are implied and are hence not specified in the instruction.
#Please refer Bharat Sir's Lecture Notes for this ...

Eg: STC ; Sets the Carry Flag.

Eg: CLD ; Clears the Direction Flag.

Important points for understanding addressing modes...

- 1) Anything given in square brackets will be an Offset Address also called Effective Address.
- 2) MOV instruction by default operates on the Data Segment; unless specified otherwise.
- 3) BX and BP are called Base Registers.
BX holds Offset Address for Data Segment.
BP holds Offset Address for Stack Segment.
- 4) SI and DI are called Index Registers
- 5) The Segment to be operated is decided by the Base Register and NOT by the Index Register.